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No. 2692:

Hotelling's Law

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by **Andrew Boyd**

Today, we meet in the middle. The University of Houston's College of Engineering presents this series about the *machines* that make our civilization run, and the *people* whose ingenuity created them.

Harold Hotelling was an accomplished economist. Trained in mathematics, he participated in the early twentieth century movement to mathematize economics. Yet, he's perhaps best known for some simple yet profound observations, one of which is now known as *Hotelling's Law*.

Hotelling's Law can be illustrated with an example. Imagine a stretch of beach a mile long on which two ice cream vendors want to sell ice cream. The flavors they offer and the prices they charge are the same, so sunbathers go to the closest cart. The question for the two vendors is, "where should they set up their carts to get the most customers?"

As objective, outside observers, we can reason that the best solution is for the carts to be stationed a quarter of a mile from opposite ends of the beach. That way, no one has to walk more than a quarter of a mile to get their ice

cream. People in the middle will be will be indifferent, since it's a quarter of a mile to either cart.



But now let's step into the shoes of one of the vendors. We see that we're doing well, serving half of the beach. But what if we rolled our cart a little toward the center of the beach? All of the people on our end won't change their behavior — we're still the closest ice cream cart. But now we're closer to the middle of the beach than our competitor, so those people in the middle are no longer indifferent. They'll come to us. In fact, the closer we move toward our competitor, the more customers we'll get.

Of course, our competitor is thinking the same thing. So where do we eventually wind up? It's not hard to see that unless the two carts are situated side by side in the middle of the beach, there's economic incentive for one or the other to move. The middle is an equilibrium point — neither vendor can benefit by moving.



In the end, by moving to the middle the vendors haven't gained anything — they're still both serving just half of the beach. But look what's happened to the sunbathers. Now a lot of people have to walk more than a quarter of a mile. Some have to walk as far as half a mile.

Hotelling's Law states that there's a natural tendency for competitors to be pulled toward a common middle ground. Ever noticed how refrigerators in the same price range look almost identical from brand to brand? Or why car dealerships all tend to pop up along the same stretch of road? Or even why the menus at McDonalds and Burger King are hard to tell apart.



One of the most commonly pointed to examples of Hotelling's Law arises in politics. When candidates position themselves too far left or right, they can lose the vote of centrists. So there's a natural pull toward the middle of the political spectrum. It's not a hard and fast rule. But it's certainly something to think about.

I'm Andy Boyd at the University of Houston, where we're interested in the way inventive minds work.

(Theme music)

Notes and references:

For a related episode, see **ECONOMICS OF ENVIRONMENT**.

There are many good references available describing Hotelling's Law, which is also known as *Hotelling's Principle*, the *Principle of Minimum Differentiation*, and *Hotelling's Linear City Model*. See, for example, D. B. Ridley, *Hotelling's Law*. From the Web site: <http://faculty.fuqua.duke.edu/~dbr1/research/Hotellings-Law.pdf>. Accessed February 23, 2011.

H. Hotelling, 1929. 'Stability in competition.' *Economic Journal*, vol. 39, 41-57.

The picture of the GE and Kenmore refrigerators are from the Web sites of these companies. All other pictures are by E. A. Boyd.

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